

Reinforcement Learning

Assignment 1

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1 Assignment 1: Reinforcement Learning

1.1 Question 1

Q1: “Prove that the policy iteration algorithm converges to the optimal policy for a finite MDP with a finite action space for each state.”

First of all, its good to define each key word:

Optimal Policy: An optimal policy is a strategy or decision-making rule that maximizes the expected cumulative reward for an agent in a given environment. An optimal policy is one that leads to the highest possible long-term reward for the agent, considering the dynamics of the environment and the available actions.

Finite MDP: A Markov Decision Process (MDP) is used to model the decision-making problems where an agent interacts with an environment over discrete time steps. The keyword "finite" only means that the MDP consists of a limited set of states and actions with respective transition function that defines the probabilities of moving from one state to another given an action, and a reward function that assigns a numerical value to each state-action pair.

Finite Action Space: A finite action space refers to a situation in which an agent has a limited number of possible actions to choose from in each state of the environment. Meaning that, for every state in the Markov Decision Process (MDP), there is a finite set of actions that the agent can take.

Policy iteration contains a few different steps:

1. **Initialization:** Start with an arbitrary policy π_0 and initialize the value function $V_0(s)$ for all states s .
2. **Policy Evaluation:** For the current policy π_k , compute the value function $V_k(s)$ for all states s using the Bellman equation:

$$V_k(s) = \sum_{a \in A} \pi_k(a|s) \left[R(s, a) + \gamma \sum_{s' \in S} P(s'|s, a) V_k(s') \right]$$

3. **Policy Improvement:** Update the policy π_{k+1} by choosing the action that maximizes the expected value for each state:

$$\pi_{k+1}(s) = \arg \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} P(s'|s, a) V_k(s') \right]$$

1.2 Question 2

Q2: “Prove that the Bellman optimality equation has a unique solution for a finite MDP with a bounded reward function.”

Setup

Consider a finite MDP with state space $\mathcal{S}$, finite action sets $\mathcal{A}(s)$, transition probabilities $P(s' | s, a)$, discount factor $\gamma \in [0, 1)$, and a *bounded* reward function: there exists $R_{\max} < \infty$ with $|R(s, a, s')| \leq R_{\max}$ for all s, a, s' .

Since $\mathcal{S}$ is finite, a value function is just a vector $V \in \mathbb{R}^{|\mathcal{S}|}$. We measure distance between two value functions with the **max-norm**:

$$\|V - U\|_{\infty} = \max_{s \in \mathcal{S}} |V(s) - U(s)|.$$

This space is *complete* (finite-dimensional real vector spaces always are).

Define the **Bellman optimality operator** T , which takes a value function V and produces a new one:

$$(TV)(s) = \max_{a \in \mathcal{A}(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V(s')].$$

A solution to the Bellman optimality equation is exactly a **fixed point** of T , i.e. a V^* with $TV^* = V^*$.

Step 1: T is a contraction

Theorem 1.1 For any two value functions V, U ,

$$\|TV - TU\|_{\infty} \leq \gamma \|V - U\|_{\infty}.$$

Proof. Fix a state s . Both $(TV)(s)$ and $(TU)(s)$ are maxima over the same finite action set, so their difference is bounded by the largest per-action gap:

$$|(TV)(s) - (TU)(s)| \leq \max_a \left| \sum_{s'} P(s' | s, a) \gamma (V(s') - U(s')) \right|.$$

(Intuitively: whichever action attains the max for V can be no more than as far “ahead” of U ’s best action as the two functions ever differ, and vice versa the reward terms are identical for V and U and cancel out.)

Since $P(\cdot \mid s, a)$ is a probability distribution, this average is at most γ times the worst-case disagreement between V and U :

$$\left| \sum_{s'} P(s' \mid s, a) \gamma (V(s') - U(s')) \right| \leq \gamma \sum_{s'} P(s' \mid s, a) |V(s') - U(s')| \leq \gamma \|V - U\|_{\infty}.$$

This bound holds for every state s , so taking the max over s gives

$$\|TV - TU\|_{\infty} \leq \gamma \|V - U\|_{\infty}.$$

□

Since $\gamma < 1$, this says T strictly shrinks distances exactly like the photocopier at $\gamma \times 100\%$ zoom.

Step 2: Contractions have a unique fixed point

Theorem 1.2 – Banach Fixed-Point Theorem If T is a contraction on a complete space (i.e. $\|TV - TU\| \leq \gamma \|V - U\|$ for some $\gamma < 1$), then T has **exactly one** fixed point, and repeatedly applying T from any starting point converges to it.

We've shown T satisfies exactly this hypothesis. So this theorem hands us existence, uniqueness, and convergence all at once, with no extra work.

A direct one-line uniqueness check (optional, self-contained)

Suppose V_1 and V_2 are both solutions: $TV_1 = V_1$ and $TV_2 = V_2$. Then by Step 1,

$$\|V_1 - V_2\|_{\infty} = \|TV_1 - TV_2\|_{\infty} \leq \gamma \|V_1 - V_2\|_{\infty}.$$

Since $\gamma < 1$, the only number $d \geq 0$ satisfying $d \leq \gamma d$ is $d = 0$. So $\|V_1 - V_2\|_{\infty} = 0$, meaning $V_1 = V_2$.

Conclusion

Theorem 1.1 For a finite MDP with bounded rewards and $\gamma \in [0, 1)$, the Bellman optimality equation has a unique solution V^* .

Proof. T is a contraction (Step 1) on a complete space, so by the Banach Fixed-Point Theorem (Step 2) it has exactly one fixed point i.e. the Bellman optimality equation has exactly one solution. □

Bonus: this same argument is why *value iteration* works. Value iteration is literally $V_{k+1} = TV_k$ repeatedly photocopying. Banach's theorem guarantees this converges to V^* no matter what V_0 you start from.

This answer is partially inspired by the explanation in [1].

1.1 Question 3

Q3: “Analyze the challenges of using linear function approximation in value-based reinforcement learning. Discuss the potential issues that may arise and propose methods to mitigate them“

2 Practical Part

Actions and rewards describing the grid-world can be found in the course book at page 60.

References

- [1] P. Salunkhe, “Understanding bellman operators: The engine of dynamic programming.” <https://medium.com/@salunkheprathmesh0/understanding-bellman-operators-the-engine-of-dynamic-programming-96eb9cb06e93>, 2023. Accessed: 2026-09-15.